Test of Independence (Chi-Square Test) Transcript

Hello everyone, welcome to lesson 2 video 2. In this video, we will discuss the chi-square test of independence. This test allows us to check whether two categorical variables are related or they are not right. Hello everyone, so far our hypothesis tests have been focused on single variable, either a mean or a proportion.

Today we are going to ask more complex and more interesting questions. Are the two variables related or not? So in this entire video, we will learn how to analyse the relationship between two categorical variables using one of the most versatile tools in statistics, which is called as chi-square test of independence. Now the fundamental question we are trying to answer is about association.

Does knowing a person's preference of one thing gives us a hint about their preference about another thing? For example, is there a association between students' home campus planning? You see here, right? Is there a relationship between a person's preferred bits planning campus and their major that they choose? Or is there a relationship between the age groups and the kind of movies or the genres that people prefer? See both of these variables are categorical because age group is bind 18 to 25, 26 to 40 and so on. And the type of movie or the genre of the movie are also categorical, right? Action, comedy, drama. Now the kind of tool that we use for testing all of these questions is called as chi-square test of independence and we'll look into it in more details.

Okay. So the logic of chi-square test is beautifully simple. It is a comparison between what we actually see and what actually we want to see.

It's a comparison between observed versus expected counts. So when I say observed versus expected count, what I mean is we perform a test looking into what we have got from our study observed count and what it should be given the null hypothesis should have been true. Okay.

So that's what is called as the chi-square test. It's a simple formulation is based on the difference between observed data and what you expect given the null hypothesis true divided by or scaled by the E value or the expected value. Okay.

This is what the entire formulation is. And for each and every observed value, we will take the difference, we'll sum it over all the data that we have. And if our chi-square value is large enough, we say that there is a relationship which exists.

If our chi-square value is small, we say that the relationship does not exist. Okay. Now again, it's a four-step process, four-step framework.

First we state the null hypothesis saying that there is no dependence or no association. The second we say that there is dependence or there is association. Then we calculate chi-square statistics using the formula that I have showed you.

And then we find the value of p or we find the value of chi-square. If the chi-square is large, it means the value of p is small. If p is small, it means p is less than, let's say if the level of significance is 5% and the p value is lesser than 0.05, then we say that there is association.

That's what is highlighted here. Right. Now, this is one of the examples, let's say the student from Pilani campus and the branches that they enjoy and they do not enjoy, let's say CS from the home branch of Pilani, 40 people enjoy it, 40 do not enjoy it, 60 MEC enjoy and 60 do not.

And then you have in total 100, right. So this is the observed data. Now, expected counts are on the right hand side.

So you are trying to understand whether there is association between the branch that students take and the home campus, right. It could be Pilani or it could be any other campus. Now if you see here, for when I change the values right now, currently the chi-square value is huge, which means that the p value would be low.

It means that there is association because we see that more computer science students, they do not enjoy, sorry, more students, they do not enjoy computer science. Okay. They enjoy MEC more.

Similarly, if I change the value of computer science, right now also this chi-square value is high, right, because people enjoy computer science more than MEC. So that's, that is what happens. This is your observed data.

This is what it should be when there should, there is no dependence because they are equally distributed, let's say between enjoyment and not enjoyment, the distribution is equal 40-40-60-60. But when you get a data which is skewed towards one class, then you test for whether the observations that you are looking for or you are looking at, is it really significant or it is just a chance, right? But right now if you see, when we calculate chi-square statistics, it came out to be really high and p value is small and hence we make that assumption that they are related. So this is what we studied.

This is very important, one of the most important test that we use for judging the association between two categorical variables. Now we have moved from judging the association between numeric variables to categorical variables and one of the most important test that we use is chi-square test. So I hope this was good for all of us to understand and ponder over what exactly is chi-square test.

In summary, chi-square test allows us to test any of those dependence-independence between two or more categorical data.

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